

SEWON PARK

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EDUCATION

University of Seoul – B.Eng. in Transportation Engineering Mar 2021 – Feb 2027 (expected)
GPA: 4.30/4.50 (98/100) · National Science & Engineering Scholarship (full tuition + stipend)
Exchange at Rikkyo University, Sociology (Mar 2025 – Aug 2025)
Coursework: Optimization, Engineering Math, Applied Statistics, Traffic Flow Theory (stochastic processes & queues)

SKILLS

Programming & Infra	Python, R, MATLAB, SQL, OR-Tools, Git, PostgreSQL, GCP
ML	XGBoost, PyTorch, scikit-learn, pandas, Bayesian, Kalman filter, SHAP, Optuna
Finance	Factor models, time-series econometrics, fixed income, portfolio optimization
AI	Fine-Tuning, Prompt Engineering, AI Agents
Languages	Japanese (JLPT N1), Korean (native), English (fluent)

PROJECTS

ProbeDensity: Traffic Density Estimation from Probe Vehicles [GitHub](#)

- Built smartphone-deployable density estimator via 31 car-following features; trained on 50K SUMO scenarios (~208K samples), benchmarked 6 model families with Optuna hyperparameter search
- Achieved 5-probe Bayesian ensemble MAE 2.18 veh/km/lane, $R^2=0.951$ (−16% vs single probe); implemented Kalman sensor fusion, grid-indexed GIS matching, and real-time monitoring across 2.2K Seoul arterial links
- Deployed FastAPI on GCP Cloud Run with Docker, Kafka, async PostgreSQL, and 145+ tests

LLM-Guided Synthetic Driving Scenario Generation [GitHub](#)

- Fine-tuned **gpt-4.1-mini** on 2,450 Seoul traffic pairs to extract 8 driving parameters; MAPE 51.5% → 10.6% via held-out evaluation, with 0% JSON format errors from constrained prompt design
- Built an 11-tool ReAct agent with hierarchical role-separated design and SQLite-backed correction-to-retraining pipeline; deployed on GCP Cloud Run with observability framework (\$0.001/request, 235x cheaper than Opus)

EXPERIENCE

CnerG	Apr 2026 – Present
<i>Analyst Intern, Global Market Team</i>	<i>Seoul, Korea</i>

- Calibrating multi-factor Python/ML pricing models for OTC-traded RECs and voluntary carbon credits; engineered features (vintage, generation method, plant age, country) from public + internal trade data
- Shipped bilingual pre-call market intelligence automation and weekly research notes (trade flows, regulatory updates) serving 35 traders across APAC/EMEA/Americas; scoped with traders, iterated on feedback, saves 45–60 min/day
- Analyzing price dynamics in illiquid, pink-sheet-style OTC environmental commodity markets across jurisdictions; designing market ontology to standardize cross-registry pricing inputs

Arthur D. Little	Jan 2026
<i>Strategy Consulting Intern</i>	<i>Tokyo, Japan</i>

- Proposed a ¥1T acquisition framework for a major automaker; mapped autonomous mobility platform risks
- Built valuation cases for 3 M&A targets using PER/EBITDA multiples, DCF, and revenue-based approaches
- Led platform-dependency analysis; presented findings in Japanese to Principals

ADDITIONAL INFORMATION

Military: ROKAF Intelligence Command (2022–2023); aide to general officer, personnel management.
Research: UOS Transportation Planning Lab (2024); ML for Seoul govt, 10 universities + 18 corporate partners.
Markets: Long-term passive investor (S&P 500 ETF); tracks BOJ/FOMC, rates, and macro indicators daily.